

MCMC3001 ENVIRONMENTAL ENGINEERING (3-0-0)

Module-2

SEWAGE:

It is a dilute mixture of the wastes of various types from the residential, public and industrial places. It includes sullage water and foul discharge from the water closets, urinals, hospitals, stables etc.

STORM WATER:

It is the surface runoff obtained during and after the rainfall which enters sewers through inlet. Storm water is not foul as sewage and hence it can be carried in the open drains and can be disposed off in the natural rivers without any difficulty.

SANITARY SEWAGE:

It is the sewage obtained from the residential buildings & industrial effluents Establishments. Being extremely foul it should be carried through underground conduits.

DOMESTIC SEWAGE:

It is the sewage obtained from the lavatory basins, urinals & water closets of houses, offices & institutions. It is highly foul on account of night soil and urine contained in it. Night soil starts putrefying & gives offensive smell. 3 It may contain large number of bacteria due to the excremental wastes of patients. This sewage requires great handling & disposal.

INDUSTRIAL SEWAGE:

It consists of spent water from industries and commercial areas. The degree of foulness depends on the nature of the industry concerned and processes involved.

SEWERAGE SYSTEM:

The entire system of collection of sewage and conveyance of sewage to treatment units by using sewers is called sewerage system. The sewerage system carries two types of flow.

DRY WEATHER FLOW (DWF):

Domestic sewage and industrial sewage collectively, is called as DWF. It does not contain storm water. It indicates the normal flow during dry season.

WET WEATHER FLOW (WWF):

The flow of storm water is called WWF.

SULLAGE:

It is defined as the waste water from domestic buildings especially from bathrooms, kitchens etc. It is merely waste water and does not create any bad smell or odors.

SLUDGE:

It is the organic matter deposited in the sedimentation tank during treatment.

SEWERAGE:

The entire system of collecting, carrying & disposal of sewage through sewers is known as sewerage.

SEWERS:

Sewers are underground pipes which carry the sewage to a point of disposal.

TRUNK SEWERS:

A trunk sewer is a one which collects the waste water or sewage from two or more main sewers.

MAIN SEWERS:

A main sewer is a one which collects the sewage from two or more sub main sewers.

SUB MAIN or BRANCH SEWERS:

A sub main or branch sewer is a one which collects the sewage from lateral sewers and discharges into main sewer.

LATERAL SEWER:

The sewer collecting the sewage directly from the building and discharging to the branch sewer is called lateral sewer.

COMBINED SEWER:

A sewer carries domestic sewage and storm water is called combined sewer.

SEWERAGE SYSTEMS

- Separate system of sewage
- Combined system of sewage
- Partially combined or partially separate system

SEPARATE SYSTEM OF SEWERAGE:

In this system two sets of sewers are laid. The sanitary sewage is carried through sanitary sewers while the storm sewage is carried through storm sewers. The sewage is carried to the treatment plant and storm water is disposed of to the river.

ADVANTAGES:

Size of the sewers are small Sewage load on treatment unit is less Rivers are not polluted Storm water can be discharged to rivers without treatment.

DISADVANTAGES:

Sewerage being small, difficulty in cleaning them. Frequent choking problem. System proves costly as it involves two sets of sewers. The use of storm sewer is only partial because in dry season the will be converted in to dumping places and may get clogged.

COMBINED SYSTEM OF SEWAGE:

When only one set of sewers are used to carry both sanitary sewage and surface water. This system is called combined system. Sewage and storm water both are carried to the treatment plant through combined sewers.

ADVANTAGES:

Size of the sewers being large, chocking problems are less and easy to clean. It proves economical as 1 set of sewers are laid. Because of dilution of sanitary sewage with storm water nuisance potential is reduced.

DISADVANTAGES:

Size of the sewers being large, difficulty in handling and transportation. Load on treatment plant is unnecessarily increased. It is uneconomical if pumping is needed because of large amount of combined flow. Unnecessarily storm water is polluted.

PARTIALLY COMBINED OR PARTIALLY SEPARATE SYSTEM:

A portion of storm water during rain is allowed to enter sanitary sewer to treatment plants while the remaining storm water is carried through open drains to the point of disposal.

ADVANTAGES:

The sizes of sewers are not very large as some portion of storm water is carried through open drains. Combines the advantages of both the previous systems. Silting problem is completely eliminated. The problem of disposing storm water from houses is simplified.

DISADVANTAGES:

The quantity of storm water admitted in sewer may increase the load on pumping and treatment units. The velocity of flow is low in dry weather.

SUITABLE CONDITIONS FOR SEPARATE SEWERAGE SYSTEMS: -

A separate system would be suitable for use under the following situations: Where rainfall is uneven. Where sanitary sewage is to be pumped. The drainage area is steep, allowing to runoff quickly. Sewers are to be constructed in rocky strata. The large combined sewers would be more expensive.

SUITABLE CONDITIONS FOR COMBINED SYSTEM: -

Rainfall is even throughout the year. Both the sanitary sewage and the storm water have to be pumped. The area to be sewered is heavily built up and space for laying two sets of pipes is not enough. Effective or quicker flows have to be provided.

METHODS OF COLLECTION

The sanitation of a town or city is done by two methods:

- Conservancy system
- Water Carriage system

SEWERAGE SYSTEMS:

CONSERVANCY SYSTEM	WATER CARRIAGE SYSTEM
Very cheap in initial cost.	It involves high initial cost.
Due to foul smells from the latrines, they are to be constructed away from living room so building cannot be constructed as compact units.	As there is no foul smell latrines remain clean and neat and hence are constructed with rooms, therefore buildings may be compact.
The aesthetic appearance of the city cannot be improved	Good aesthetic appearance of city can be obtained.
For burial of excremental matter large area is required.	Less area is required as compared to conservancy system.
Excreta is not removed immediately hence its decomposition starts before removal,	Excreta are removed immediately with water, no problem of foul smell or hygienic trouble.
This system is fully depended on human agency .In case of strike by the sweepers; there is danger of insanitary conditions in	As no human agency is involved in this system ,there is no such problem as in case of conservancy system

SOURCES OF SEWAGE

SANITARY SEWAGE IS PRODUCED FROM THE FOLLOWING SOURCES:

When the water is supplied by water works authorities or provided from private sources, it is used for various purposes like bathing, utensil cleaning, for flushing water closets and urinals or washing clothes or any other domestic use. The spent water for all the above needs forms the sewage. Industries use the water for manufacturing various products and thus develop the sewage. Water supplied to schools, cinemas, hotels, railway stations, etc., when gets used develops sewage. Ground water infiltration into sewers through loose joints. Unauthorized entrance of rain water in sewer lines.

QUANTITY OF SANITARY SEWAGE AND STORM WATER

The determination of sanitary sewage is necessary because of the following factors which depend on this:

- ❖ To design the sewerage schemes as well as to dispose a treated sewage efficiently.
- ❖ The size, shape and depth of sewers depend on quantity of sewage.
- ❖ The size of pumping unit depends on the quantity of sewage.

FACTORS AFFECTING THE QUANTITY OF SANITARY SEWAGE FLOW: -

- ❖ Rate of water supply
- ❖ Population
- ❖ Type of area served as residential, industrial or commercial Infiltration and exfiltration

RATE OF WATER SUPPLY:

The rate of sewage may be 60 to 70 percent of water supply due to various reasons such as consumption, evaporation, use in industries etc. This may be changes daily, seasonal and also standard of living of people.

POPULATION:

As the population increases the quantity of sewage also increases because the consumption of water is more.

TYPE OF AREA SERVED:

The quantity of sewage depends upon the type of area as residential, industrial or commercial. The quantity is depending on population if it is residential, type of industry if it is industrial,

Commercial and public places can be determined by studying the developing of other such places.

INFILTRATION AND EXFILTRATION:

- ❖ Infiltration is the leakage of water from the ground surrounding the sewer.
- ❖ Exfiltration is the leakage of sewage from the sewer into the ground surrounding the sewer.

The quantity of water increased through infiltration depends on the following factors:

- Length of sewer
- Size of sewer
- Sub soil water head
- Nature and type of soil.

FACTORS AFFECTING THE QUANTITY OF STORM WATER FLOW: -

- ❖ Intensity of rainfall
- ❖ Catchment areas
- ❖ Duration of rainfall

ESTIMATION OF QUANTITY OF STORM WATER: -

- ❖ Rational method
- ❖ Empirical Method

DESIGN OF SEWERS

Average daily flow = (70 – 80) % average water consumption

i.e. Average Daily Flow (ADF) of Sewage = 75%

Maximum Daily Flow = 1.5 x ADF

DESIGN PERIOD

The future period for which the provision is made in designing the capacities of the various components of the sewerage scheme is known as the design period.

The design period depends upon the following:

- Ease and difficulty in expansion.
- Amount and availability of investment.
- Anticipated rate of population growth, including shifts in communities, industries and commercial investments.
- Hydraulic constraints of the systems designed, and

- Life of the material and equipment.

Following design period can be considered for different components of sewerage scheme:

- ❖ Trunk or main sewers: 40 to 50 years
- ❖ Treatment Units: 15 to 20 years
- ❖ Pumping plant: 5 to 10 years

FLUCTUATIONS IN SEWERS

The sewage flow, like the water supply flow, is not constant in practice but varies. The fluctuation may, in a similar way, be seasonal or monthly, daily and hourly. Fluctuation in flow occurs from hour to hour and from season to season.

The seasonal variations are due to climatic effect, more water being used in summer than in winter. The daily fluctuations are the outcome of certain local conditions, involving habits and customs of people. In India, however, Sundays or other holidays involve activities which permit greater use of water. Hourly variations are because of varying rates of water consumption in different hours of the day.

The first peak flow generally occurs in the late morning it is usually about 200 percent of the average flow while the second peak flow generally occurs in the early evening between 6 and 9 p.m. and the minimum flow occurring during the night after twelve or early hours of the morning is generally about half of the average flow.

HYDRAULICS OF SEWERS

SELF CLEANSING VELOCITY:

The sewers should be laid at such a gradient that a minimum velocity, which will prevent the silting of particles in sewers are developed. Such a minimum velocity is known as self-cleansing velocity.

Self-cleansing velocity is generally,

- ❖ 0.6 m/s for sanitary sewer
- ❖ 1m/s for storm sewer.

NON-SCOURING VELOCITY:

The upper limit of velocity is set by scouring action of sewage.

If the velocity of flow exceeds a certain limit, the particles of solid matter start to damage the inside surface of sewers or in other words, a scouring action takes place.

The maximum permissible velocity at which no such scouring action will occur is known as non-scouring velocity and it mainly depends on the material of sewers.

Generally,

- ❖ Sanitary sewer = 2.4 m/sec
- ❖ Storm sewer = 3 m/sec

SHAPES OF SEWERS

- Circular sewer section
- Non circular sewer section

CIRCULAR SEWER SECTION:

Sewers are generally circular in shape. The sewers are running at least half full.

If the depth in a circular sewer is less than one-half, there is considerable reduction in velocity and discharge and it results in poor performance.

ADVANTAGES OF CIRCULAR SEWERS:

- The perimeter of circular sewer is the least with respect to the sewer of other shape.
- The inner surface is smooth hence the flow of sewage is uniform and there is no chance of deposition of suspended particles.
- From hydraulic point of view, circular section provides more hydraulic mean depth. Construction cost and material requirement is less.

Non-circular shaped sewers are also adopted for the following reasons:

- They can be construct in such a convenient shape and size so that a man can enter the sewer for cleaning, maintenance, etc.
- The process of construction is easy.
- The structural strength is more.
- Cost of construction is low.

NON-CIRCULAR SEWER SECTIONS:

- ❖ Standard Egg Shape
- ❖ New Egg Shape
- ❖ Horse Shoe Shape
- ❖ Parabolic Shape
- ❖ Semi Elliptical Shape
- ❖ Rectangular Shape
- ❖ U- Shaped
- ❖ Semi Circular Shape
- ❖ Basket Handle Shape

STANDARD EGG SHAPE

- It is preferred for combined sewers.
- Its advantage over circular shape is that it gives higher velocity during low flow of same capacity.
- It is difficult to construct and less stable.
- It requires additional bedding of brick masonry or concrete to make it stable.

HORSE SHOE SHAPED SEWER:

- This may be used for large sewers with heavy discharges such as for trunk and outfall sewer. 15
- Such sewer is suitable when headroom for the construction of sewer is limited.
- The invert of this section may be flat, parabolic or circular.
- Its height is more than its width.
- Its wall most inclined with semicircular arch at top.

PARABOLIC SHAPED SEWER:

- The upper arch of the sewer forms the shape of parabola.
- This may be used for carrying comparatively small quantities of sewage.
- The invert may be elliptical or parabolic.
- It is found to be economic in construction.

SEMI ELLIPTICAL SHAPED SEWER:

- It may be used for soil i.e. soft soil as it is more stable.
- It is useful only for carrying large number or amounts of sewage.
- It is adopted when sewers are greater than 1.8m in diameter

RECTANGULAR SHAPED SEWER:

- Generally, it is used for covered storm water drains.
- It is stable and easy to construct.
- Sometimes it is used to work as a storage tank.

U SHAPED SEWER:

- The section may have true shape of letter 'U', or a small trench of
- U-Shape can be set up in the large section.
- That trench is known as cunette.
- Such type of sewer used for combined sewer having maximum flow of storm water.
- It is used for longer sewers and especially in open cuts.

SEMI CIRCULAR SHAPED SEWER:

- This section gives a wider base at bottom and hence it becomes suitable for constructing large sewers with less available headroom.
- It is out dated.

BASKET HANDLE SHAPED SEWER:

- In this type of sewer, the upper portion of sewer has got the shape of a basket- handle.
- The bottom portion is narrower in width than the upper portion.
- It carries small discharge through the bottom narrow portion and during monsoon it runs full.
- It is also out dated.

FACTORS CONSIDERED WHILE SELECTING MATERIAL FOR SEWER

- ❖ Resistance to corrosion
- ❖ Resistance to abrasion
- ❖ Strength and durability
- ❖ Weight of the material
- ❖ Imperviousness
- ❖ Economy and cost
- ❖ Hydraulically efficient

RESISTANCE TO CORROSION:

Sewer carries wastewater that releases gases such as H_2S . This gas in contact with moisture can be converted into sulfuric acid. The formation of acids can lead to the corrosion of sewer pipe. Hence, selection of corrosion resistance material is must for long life of pipe.

RESISTANCE TO ABRASION:

Sewage contains considerable number of suspended solids, part of which are inorganic solids such as sand or grit. These particles moving at high velocity can cause wear and tear of sewer material. This abrasion can reduce thickness of pipe and reduces hydraulic efficiency of the sewer by making the interior surface rough.

STRENGTH AND DURABILITY:

The sewer pipe should have sufficient strength to withstand all the forces that are likely to come on them. Sewers are subjected to considerable external loads of backfill material and traffic load, if any. They are not subjected to internal pressure of water. To withstand external load safely without failure, sufficient wall thickness of pipe or reinforcement is essential. In addition, the material selected should be durable and should have sufficient resistance against natural weathering action to provide longer life to the pipe.

WEIGHT OF THE MATERIAL:

The material selected for sewer should have less specific weight, which will make pipe light in weight. The lightweight pipes are easy for handling and transport.

IMPERVIOUSNESS:

To eliminate chances of sewage seepage from sewer to surrounding, the material selected for pipe should be impervious.

ECONOMY AND COST:

Sewer should be less costly to make the sewerage scheme economical.

HYDRAULICALLY EFFICIENT:

The sewer shall have smooth interior surface to have less frictional coefficient.

SEWER MATERIALS

- ❖ Asbestos Cement (AC) Sewer
- ❖ Brick Sewer
- ❖ Cement Sewer
- ❖ Cast iron (C.I) Sewer
- ❖ Steel Sewers
- ❖ Plastic Sewers
- ❖ Vitrified Clay or Stoneware Sewers
- ❖ Lead Sewers

DESIGN CRITERIA FOR SEWER SYSTEM

For the design of sewer, the following factors are to be considered.

- Rate of water supply
- Intensity of rainfall
- Type of sewer material
- Velocity of flow
- Slope of the ground
- Shape of the sewer.

VELOCITY OF FLOW

The velocity of flow in a sewer line is determined by the following formulae:

- ❖ Chezy's Formula
- ❖ Kutter's Formula
- ❖ Basin's Formula
- ❖ Manning's Formula
- ❖ Crimp and Burge's Formula
- ❖ Hazen's Williams Formula

CHEZY'S FORMULA:

$$V = C \sqrt{mi} \text{ m/sec.}$$

Where, V – Velocity of flow in m/sec.

C – Chezy's Constant

m – hydraulic mean depth (A/P) in m.

i - hydraulic gradient or hydraulic slope

KUTTER'S FORMULA:

According to this formula, value of Chezy's constant (C) is given by,

$$C = \frac{23 + \frac{0.00155}{i} + \frac{1}{N}}{1 + \left(23 + \frac{0.00155}{i}\right) \frac{N}{\sqrt{m}}}$$

BASIN'S FORMULA:

$$C = \frac{157.6}{[1.81 + (K/R^{1/2})]}$$

MANNING'S FORMULA:

$$v = \frac{1}{n} m_3^2 i_2^1$$

Where, n – manning's constant.

m – hydraulic mean depth (A/P) in m.

i - hydraulic gradient or hydraulic slope

CRIMP AND BURGE'S FORMULA

$$V = 83.50 m^{2/3} i^{1/2}$$

Where,

m – hydraulic mean depth (A/P) in m.

i - hydraulic gradient or hydraulic slope

HAZEN'S WILLIAMS FORMULA:

$$V = 0.85 CH \times m^{0.63} i^{0.54}$$

Where,

CH - Hazen's Williams constant.

m – hydraulic mean depth (A/P) in m.

i - hydraulic gradient or hydraulic slope

1. Calculate the velocity of flow in a sewer of diameter 1.36 m. The gradient of sewer line is 1 in 420. If $n = 0.012$ in Manning's formula, what is the discharge when running one-half full.

2. Calculate the discharge for a stoneware sewer, running full. The diameter of the sewer is 200 mm and it is laid at a slope of 1 in 72. Take $n = 0.013$ in Manning's formula.

3. Calculate the discharge of a sewer, running full having a diameter of 200 mm and laid at a slope of 1 in 240 by using Crimp and Burge's formula.

CLEANING AND MAINTENANCE OF SEWERS

The sewers should be properly cleaned and maintained in good working condition. It should be noted that sewers which are once laid and buried into the ground should not be forgotten as they are also liable to corrosion, deterioration, erosion etc.

CAUSES

- ❖ Breakage of sewers
- ❖ Clogging
- ❖ Odours

BREAKAGE OF SEWERS:

The sewers are sometimes broken after being laid under the ground. Several factors may contribute to the breakage of sewers, being poor foundation, excessive superimposed loads, impact due to vibrations etc. The presence of corrosive matter in sewage will slowly eat away the material of sewer and it will ultimately result in the breakage of sewer.

CLOGGING:

Clogging mainly occurs in sewers of small size as it is possible for a man to enter such sewers and clean them. The clogging may be due to waste building materials, ashes, deposition of sand and grit. It may also be caused by the contribution of greasy matter from garages, hotels, soap industries. It may be noted that clogging is predominant in sewers lay at flat slopes such that the self-cleansing velocities are not developed.

ODOURS:

The organic matter present in sewer decomposes and gives out unpleasant odours.

It is essential to clean the sewers to bring down the intensity of such unpleasant odours to the minimum possible extent.

METHODS OF CLEANING OF SEWERS

- ❖ Cleaning and Flushing
- ❖ Cleaning of catch pits
- ❖ Inspection
- ❖ Periodical Repairs
- ❖ Proper Connections

CLEANING AND FLUSHING

The cleaning of large sewers is done manually. The man enters the sewers through manholes and scraps the dies of sewer by hand. The scraped material is removed through manholes. The cleaning of small sewers is affected by flushing. For this purpose, the automatic flushing tanks are sometimes installed on the sewer line.

CLEANING OF CATCH PITS:

The catch pits used to collect storm water are cleaned after every storm. The catch pits contain debris, silt, sand etc. and even the water contained in catch pits is likely to give rise to the growth of mosquitoes. A slight trace of organic matter in silt will give unpleasant odour. The oil and grease traps are also periodically cleaned to avoid the nuisance due to unpleasant odours.

INSPECTION:

The sewer and its appurtenances should be inspected at regular intervals to ascertain their proper working. The inspection includes examination of structures, measuring rate of flow, determining the amount of clogging. In case of small sewers, inspection may be also carried out through lamp holes. In case of large sewers, inspection is carried out by entering and examining the condition of sewer.

Extreme care should be taken before entering the sewer and absence of poisonous and explosive gases in sewer should be confirmed before entering the sewer. A fire-hose with nozzle may be also inserted in the sewer and the water under pressure may be discharged through the nozzle to clean the sewer.

When flushing is inadequate to remove obstructions in the sewer, the following methods are employed to make the sewer clean.

Flexible rod.

Mechanical tools.

Use of pills.

The precautions to be taken during inspection, only experienced workers should be allowed to enter the sewer for inspection. The explosion proof electric lighting equipment should only be used for inspection inside the manhole. The workers should be provided with proper protective measures such as gas masks, rubber gloves, gum boots, non-sparking tools, safety belt etc. The workers should be trained to guard against the water borne diseases through sewage. They should also be advised to keep excellent personal hygiene. The warning signs for traffic should be placed on the road on which manholes are opened out for inspection.

PERIODICAL REPAIRS:

The damaged portions of sewers should be immediately repaired. The brick sewers require frequent repairs. The broken bricks should be replacing and pointing to the brickwork should be done at regular intervals. The manholes also should be periodically examined and repaired, if necessary. The damaged or broken covers of street inlet or catch basin should be replaced. The manhole covers which have loose by traffic should be tightened. The defective connections between the house sewers and the street sewers should be immediately repaired. The ventilating columns or shafts should be checked at regular intervals and it should be ensured that they are functioning properly.

PROPER CONNECTIONS:

The connection of lateral sewers with branch sewers should be carried out by authorized licensed plumbers only. The plumbing work of house drainage should be carefully done and the joints should be made water tight.

WASTEWATER TREATMENT

OBJECTIVES

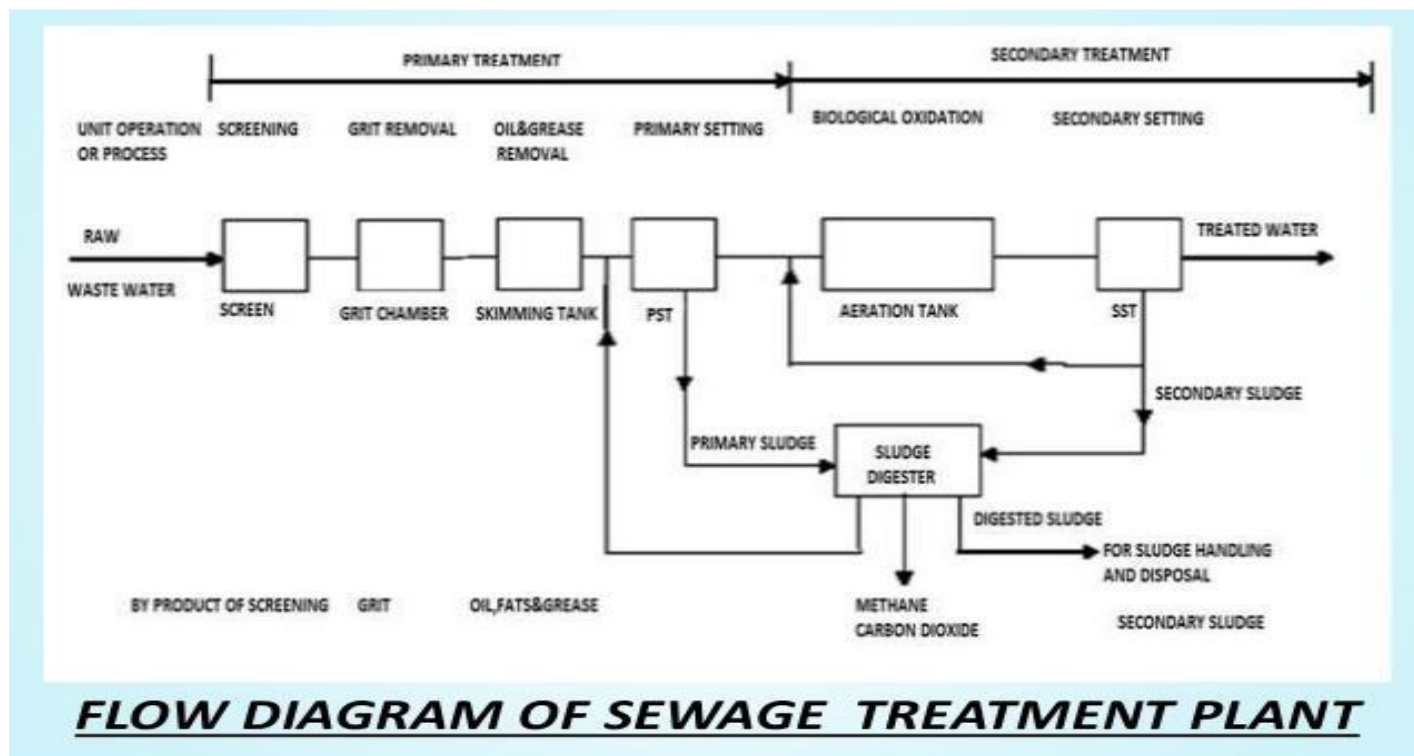
- The purpose of waste water treatment is to remove contaminant from water so that the treated water can meet the acceptable quality standard.
- The quality standard usually depends whether the water will be reused or discharged into river.
- Before disposing of sewage into river or land, sewage has to be treated to make it safe and to make it harmless.
- Methods of waste water treatment depends on composition of waste water and required quality for treated water.

- Treatment process are broadly classified as physical, chemical and biological treatments.
- Physical treatment methods utilize physical separation of pollutant such as by filtration etc.
- Chemical treatment methods utilize chemical characteristics of pollutant for purification. For e.g., Coagulation etc.
- Biological treatment methods utilize biological characteristics of pollutants such as bacteria, viruses by purification.

Other purpose of waste water treatment includes:

- ❖ To reduce strength of sewage
- ❖ To make waste water less offensive
- ❖ To prevent public health from toxic effect of pollutant
- ❖ To conserve nature

LAYOUT OF SEWAGE TREATMENT PLANT



OPERATIONS UNIT : Operations Unit are the treatment methods, in which the applications of physical forces are carried out. Example: Screening, Flocculation, Sedimentation.

PROCESSES UNIT : Unit processes are the treatment methods, in which the chemical or biological activities are involved. Example: Chemical Preparation, Gas Transfer, Adsorption, Disinfection, Combustion etc.

STEPS OF SEWAGE TREATMENT PROCESS: The four processes are:

- ❖ Preliminary Treatment
- ❖ Primary Treatment
- ❖ Secondary or Biological Treatment
- ❖ Tertiary or Advanced or Final Treatment.

PRELIMINARY TREATMENT

It is the first stage in treatment process, in which the floating materials such as tree branches, dead animals, papers, pieces of rags, wood etc. are separated from the sewage.

FUNCTIONS OF PRIMARY TREATMENT:

Removes oils and greases. Reduces the BOD of sewage about 25 %. Removes larger settleable, in-organic solids etc.

Some of the treatment technique applied for preliminary treatment purpose are:

- ❖ Screening
- ❖ Grit removal
- ❖ Skimming

SCREENING:

Screening is the first unit operation in waste water treatment plants. Screens are devices with openings of uniform size and the screening units may consists of parallel bars, rods, gratings or wire mesh or perforated plates. Screens may be of any shape and generally they are rectangular or circular.

PURPOSE: Screening is used to remove the floating matters of comparatively larger in size like plastics, clothes, dead animals from waste water. If the larger materials are not removed, they will clog or damage the valves, appurtenances, small pipes and pumps etc.

LOCATION: The screens are generally located just prior to the grit chambers. If the quality of grit is not much importance, the screens may be placed after grit chambers. Sometimes the screens may be located along with grit chambers. The screens are generally placed in inclined position with an angle of 45° (varies from 30° to 60°) with horizontal direction of flow.

GRIT BASINS or GRIT CHAMBERS:

Sewage contains both organic and inorganic materials, and grit is the heavy mineral material found in raw sewage, it may contain sand, gravel, silt, cinders, broken glass, small fragments of metal and other inorganic solids.

Grits are small, non-biodegradable particles which are heavier than suspended organic matters. Grits are removed by carefully regulating the flow velocity of sewage in grit removal tank. They reduce the excessive accumulation of grit in such units.

Grit chambers are provided to remove grit, sand and other inorganic matter from the sewage. Grit chambers are provided to protect moving mechanical equipment from abrasion. They reduce the formation of heavy deposits in channels, pipelines or conduits.

LOCATION:

Grit chambers are placed after pumping stations and before the screens.

Due to some practical considerations, grit chamber may be placed after screens also.

Skimming

Skimming is the process of removal of fatty and oily material from sewage. Fats and oils are collected at top of the liquid which are then removed by skimmers. The tank in which scum formation is carried out by diffusion of air through the sewage is called skimming tanks.

DISPOSAL OF SKIMMINGS: The oil and greasy materials removed by skimming tanks are disposed either by burning or burial. Sometimes, the disposal of skimmings can be converted into soap lubricants, candle and other non-edible products.

PRIMARY TREATMENT OF WASTEWATER

After removal of floatable solids, grits and fats, next step in treatment is removal of

remaining suspended solids as much as possible. The main objective of primary treatment is to reduce strength of sewage by removing suspended materials.

Some common technique applied for primary treatment of sewage are

- Sedimentations Tanks
- Imhoff Tanks
- Septic Tanks
- Chemical precipitation Tanks

Sedimentations Tanks

Sedimentation tank is used for removal of suspended solids and some organic matters. The process of sedimentations reduces the strength of sewage to the extent of about 30 to 35 %. The quantity of settleable solids in the sewage is reduced to the extent of about 80 to 90%.

The materials collected at the bottom of the Tank is called **sludge** and the partially treated sewage are called **Effluent**.

Sedimentation tanks are also known as settling tanks or waste water clarifiers.

SEDIMENTATION WITH COAGULATION:

Very fine suspended particles of wastewater cannot be removed by plain or primary sedimentation tank and they can be settled by increasing their size by changing them into flocculated particles.

Sedimentation with coagulation is defined as the process of removing the very fine suspended particles present in the waste water that cannot be removed by primary sedimentation by increasing their size by using chemicals.

The chemicals used for increasing the size of the particles are called **coagulants**.

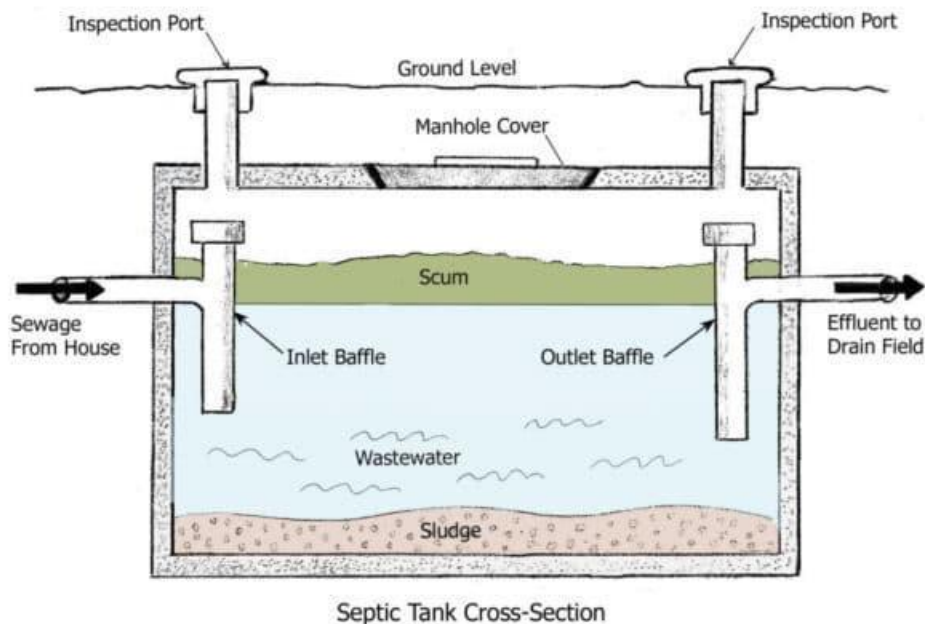
The various chemicals or coagulants are:

Alum , Ferric Chloride , Ferric Sulphate , Chlorinated Copperas.

The coagulants mixed with water forms a gelatinous precipitate called floc, which attracts the fine mud particles and other colloidal matters form a bigger sized flocculated particle.

The process of adding the chemicals is called **coagulation** and the formation of bigger sized flocculate particles is called **flocculation**.

SEPTIC TANK:



A septic tank is defined as a primary sedimentation tank with a longer detention period 12 to 36 hours with an additional provision for digestion of the settled sludge.

Due to anaerobic decomposition of the settled sludge, foul gases will be evolved in the tank and the septic tank will be completely covered and provided with a high air vent pipe for escape of gases.

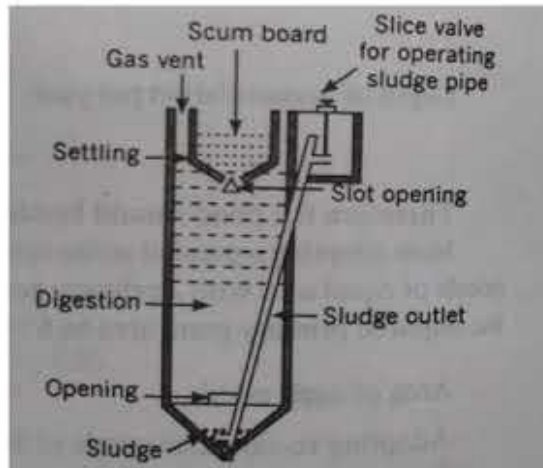
Septic tanks are provided in places, where sewer pipes have not been laid and especially septic tanks are provided for isolated communities, schools, hospitals, hotels and other isolated public institutions etc.

IMHOFF TANK:

These tanks are the improvements over the septic tanks. Karl Imhoff of Germany invented this tank. The Imhoff tank, is a two-storey septic tank, composed of an upper sedimentation compartment, and a bottom sludge digestion compartment.

The Imhoff tank is a primary treatment technology for raw wastewater, designed for solid-liquid separation and digestion of the settled sludge.

Cross section of Imhoff tank



SECONDARY TREATMENT OF WASTE WATER

The effluent from the primary sedimentation tank consists of 45 to 60 % of unstable organic matters, originally present in the sewage. The larger and medium sized solids have been removed by settling in sedimentation tanks and the organic matter present in the sewage as colloidal matter is carried away by effluent from the settling tanks.

The further treatment of sewage is called secondary treatment or biological treatment and in which the conversion of organic matter into stable forms by oxidation or nitrification.

The secondary treatment of sewage involves various methods and are classified into two important processes.

- Filtration process
- Activated-sludge process.

FILTRATION PROCESS:

In secondary treatment, the filter units consist of open beds of coarse aggregate Over which the sewage is spreaded or sprinkled intermittently. The necessary contact surface for the growth of aerobic bacteria is provided by the aggregates in the bed. By doing this, the aeration is provided by nature.

DIFFERENT TYPES OF FILTERS:

- ❖ Contact Beds
- ❖ Intermittent Sand Filters
- ❖ Trickling Filters

CONTACT BEDS

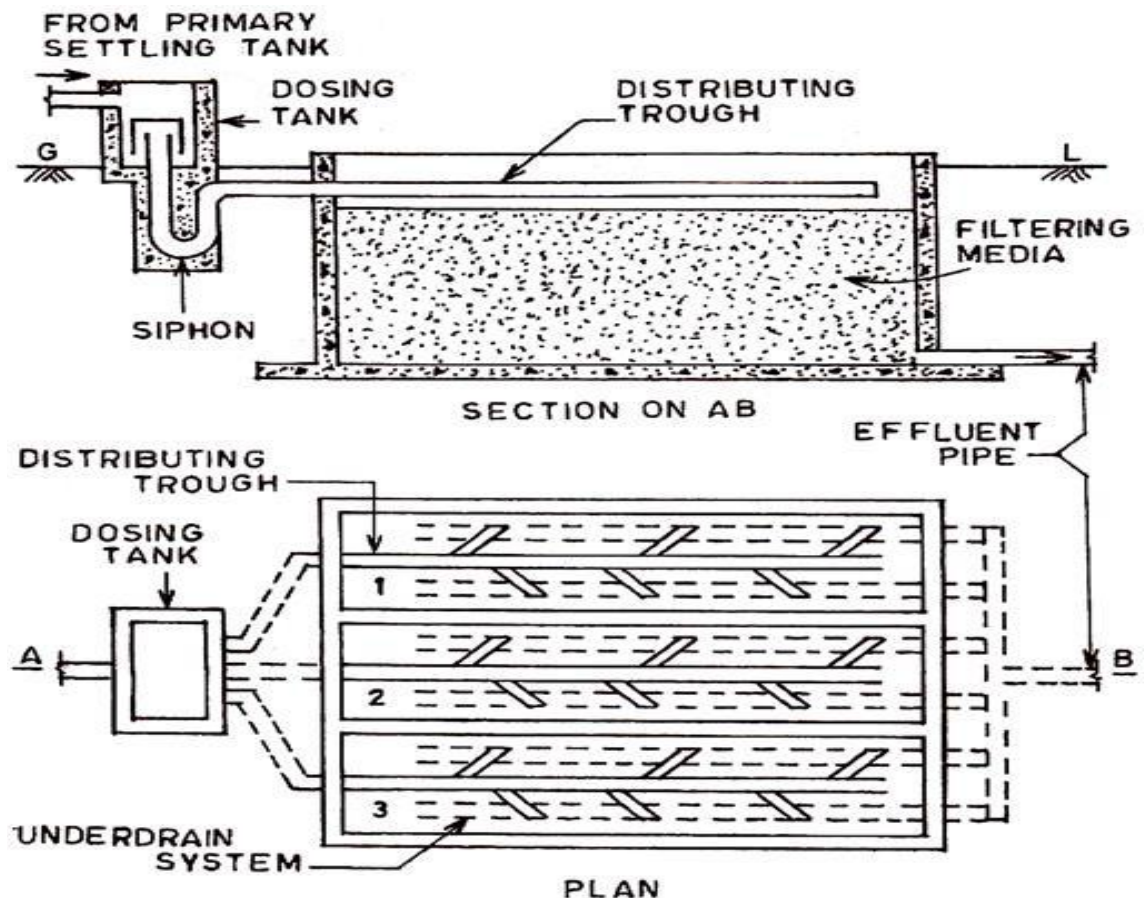


Fig. 14.2 Contact bed sewage filter.

In ancient time contact beds were very popular in the treatment of sewage, but now adays these are similar in construction to the intermittent sand filters, the only difference being in the filtering media.

The filtering media consists of 2 to 2.5 cm size broken stone ballast or brick ballast.

The depth of the filtering media is between 90-150 cm.

The sewage is uniformly applied over the whole surface of the filtering media, by means of distribution troughs and is collected at the bottom by means of a system of under drain pipes.

INTERMITTENT SAND FILTERS:

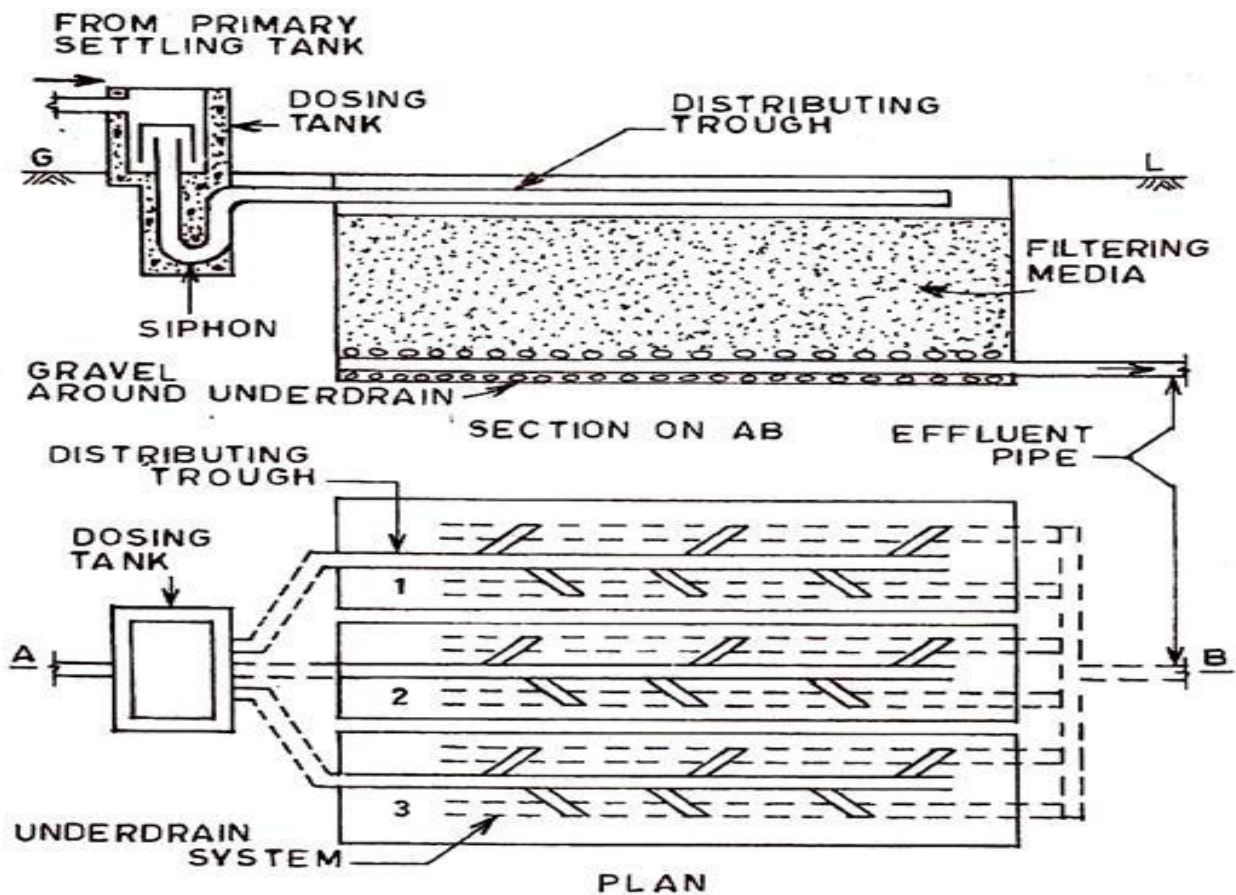
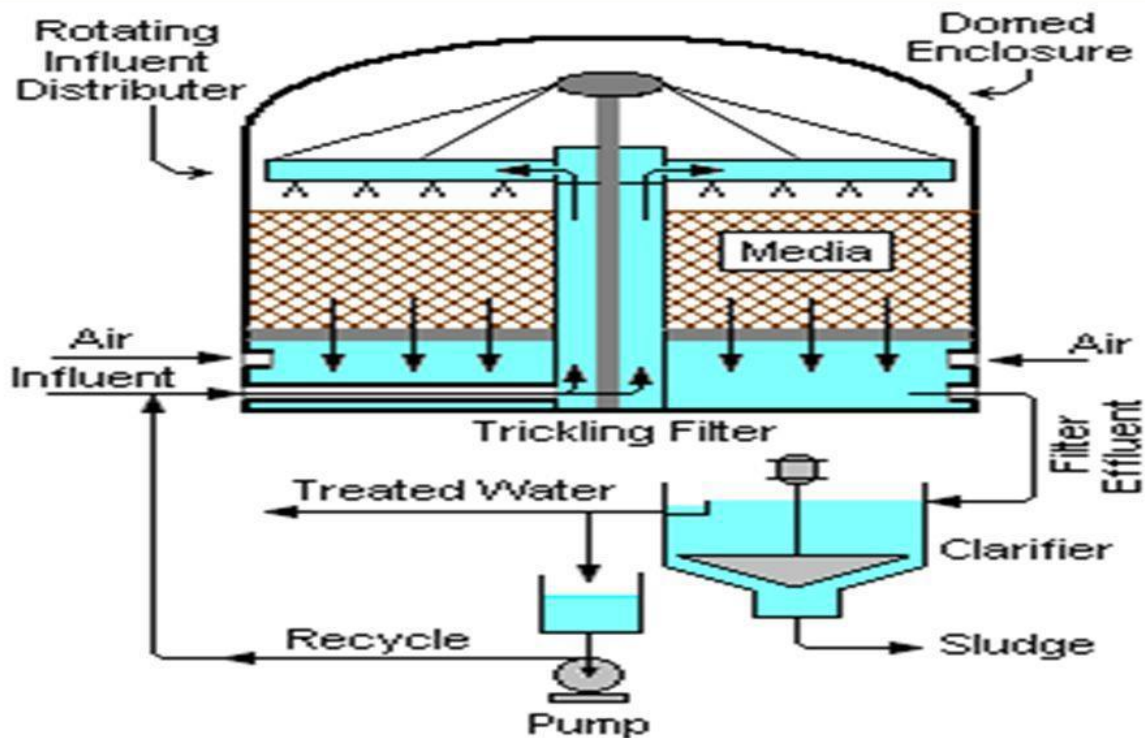


Fig. 14.1 Intermittent sand filters.

These are the early development of sewage treatment units. These are similar in Construction to the slow-sand filter of water treatment. These require larger area, due to which these are not commonly employed in modern sewage treatment works. The filter media is finer than that in the contact beds. There is no concrete lining around the filter media, as was there in contact beds.

TRICKLING FILTERS:

Trickling filter



Trickling filters (TFs) are used to remove organic matter from wastewater.

- The TF is an aerobic treatment system that utilizes microorganisms attached to a medium to remove organic matter from wastewater.
- This type of system is common to a number of technologies such as rotating biological contactors and packed bed reactors (biotowers). These systems are known as attached-growth processes.
- Trickling filter is an attached growth process i.e. process in which microorganisms responsible for treatment are attached to an inert packing material.
- Packing material used in attached growth processes include rock, gravel, slag, sand, red wood, and a wide range of plastic and other synthetic materials.
- The wastewater in trickling filter is distributed over the top area of a vessel containing non-submerged packing material.
- Air circulation in the void space, by either natural draft or blowers, provides oxygen for the microorganisms growing as an attached biofilm. During operation, the organic material present in the wastewater is metabolized by the biomass attached to the medium.

- The biological slime grows in thickness as the organic matter abstracted from the flowing wastewater is synthesized into new cellular material. The thickness of the aerobic layer is limited by the depth of penetration of oxygen into the microbial layer.
- The micro-organisms near the medium face enter the endogenous phase as the substrate is metabolized before it can reach the micro-organisms near the medium face as a result of increased thickness of the slime layer and lose their ability to cling to the media surface.
- The liquid then washes the slime off the medium and a new slime layer starts to grow. This phenomenon of losing the slime layer is called sloughing.
- The sloughed off film and treated wastewater are collected by an underdrainage which also allows circulation of air through filter.
- The collected liquid is passed to a settling tank used for solid- liquid separation.

ACTIVATED SLUDGE PROCESS

In secondary treatment, the ASP consists of aeration tanks of long detention period, in which activated sludge is mixed with sedimented sewage is agitated and aerated.

The sludge coming from the secondary sedimentation tank is called activated sludge and contains more quantity of biologically active oxygen. In the sewage treatment there are two de-composition process takes place.

- ❖ Aerobic Decomposition Process
- ❖ Anaerobic Decomposition Process

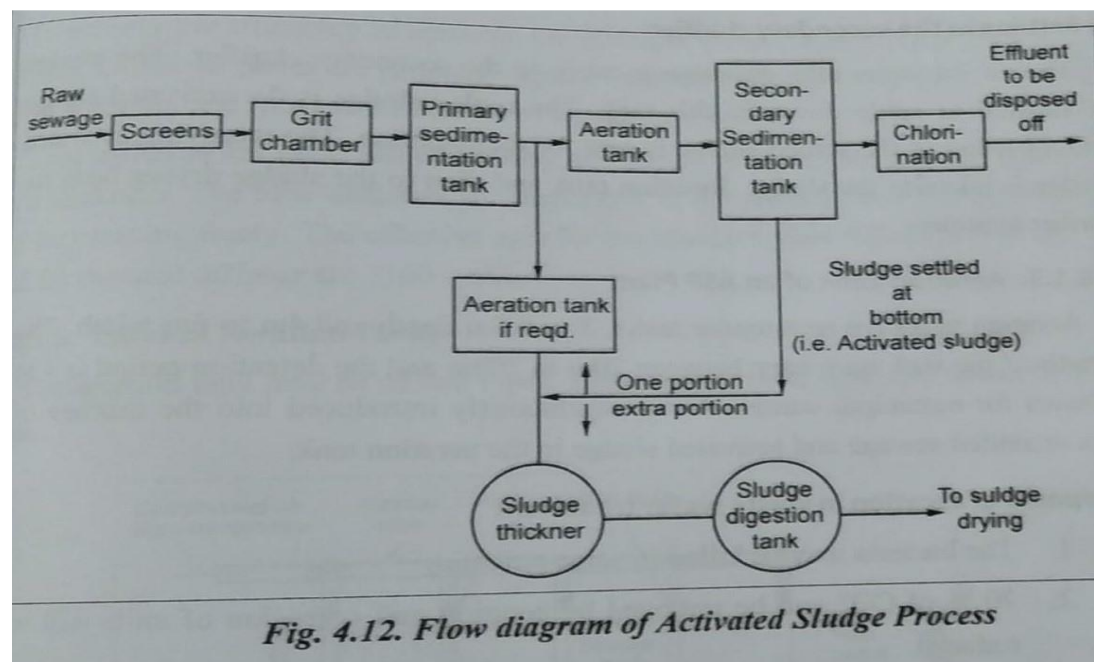


Fig. 4.12. Flow diagram of Activated Sludge Process

In secondary treatment process, only aerobic decomposition occurs and aerobic decomposition has the following advantages over anaerobic decomposition.

Aerobic decomposition does not produce bad smells and gases.

More active than anaerobic bacteria – the rate of doing work by aerobic bacteria is three times the work doing by anaerobic bacteria at 30°C.

Activated sludge process is a process for treating sewage and waste water commonly referred as effluent using bacteria (to degrade the biodegradable organics) and air (Oxygen for respiration).

Activated sludge refers to a mixture of microorganisms and suspended solids.

The sewage effluent from the primary sedimentation tank is mixed with 25 % of its own volume of activated sludge.

The activated sludge contains a larger concentration of highly active aerobic microorganism.

The mixture of sewage effluent and activated sludge enters an aeration tank, where micro-organism is mixed together with a large quantity of air, for a period of about 4 to 8 hours.

Under this condition, the micro-organisms will oxidize the organic matter and the suspended and colloidal matter tends to coagulate and form a precipitate.

This precipitate settles down in the secondary sedimentation tank instantly.

The settled sludge called activated sludge is again recycled to the head of aeration tank and again mixed with raw sewage.

Activated sludge is being produced continuously by this process and a portion of produced activated sludge is utilized in aeration tank.

The excess of activated sludge is disposed of properly along with the sludge collected during primary treatment after digestion.

TERTIARY OR FINAL TREATMENT OF WASTE WATER:

Tertiary treatment of waste water is final treatment process in which all the chemical and biological agents are completely removed from sewage before disposal into river.

After the primary and secondary treatment of sewage, the sewage is a stabilized and a harmless one. However, the primary and secondary treatments remove only conventional constituents like TSS, BOD, COD, Organic carbon. Nitrogen, phosphorus etc.

The non conventional compounds and other emerging compounds may still be present in sewage.

Also sewage contains some roundworms and bacterial pathogens that are difficult to control.

All these factors affect the public health and therefore, the sewage needs to undergo tertiary treatment before disposal or reuse for industrial, agricultural and municipal purposes.

The tertiary treatment required will be always based on the type of water reuse and degree of quality of the reclaimed water.

OBJECTIVES OF TERTIARY TREATMENT PROCESS:

- ❖ Removal of suspended solids
- ❖ Removal of dissolved solids like salts
- ❖ Removal of nitrate and phosphate
- ❖ Killing of microorganisms

REMOVAL OF SUSPENDED SOLIDS:

Suspended solids are removed by two methods:

Microstraining:

In this method, sewage is placed in rotating drum filter of pore size 25-35 μm and then drum is rotated. During rotation, clear water comes out of drum and suspended solids remains inside drum.

Chemical coagulation and filtration:

In this method, precipitating agents such as alum is added in sewage. Fine suspended solids adsorb to the surface of $\text{Al}(\text{OH})_3$ precipitate, finally precipitate with adsorbed solids are separated by filtration.

REMOVAL OF DISSOLVED SOLIDS LIKE SALTS:

Adsorption by activated carbon: Dissolved solids can be removed by filtering the water through filter containing activated carbon particle.

Reverse osmosis: Reverse osmosis removes dissolved solids like NaCl and microbial cells.

REMOVAL OF NITRATE AND PHOSPHATE:

If sewage after treatment is to be discharge into river, nitrate and phosphate should be removed from sewage before disposal.

It is because nitrate and phosphate causes eutrophication.

These plant nutrients are removed by biological process.

- ❖ At first sewage is placed in a tank containing nitrifying bacteria.
- ❖ These bacteria converts ammonium salt and nitrite into nitrate.
- ❖ Then the sewage is placed into second tank containing denitrifying bacteria.
- ❖ These bacteria converts nitrate into Nitrogen gas that leaves the sewage.
- ❖ Phosphate is also removed by bacteria by microbial assimilation process.

KILLING OF MICRO-ORGANISMS:

Finally microorganisms in sewage are killed by disinfection like chlorination.

CHLORINATION OF SEWAGE:

The process of killing the pathogenic bacteria present in the waste water is called disinfection of sewage. Chlorination is the process of treating the sewage with chlorine in order to disinfect the pathogens, either before or after the treatment. Chlorine is added to the sewage for the following purposes

- To remove the odours.
- To break down the sulphur compounds.
- To kill the bacteria cells.
- To prevent the flies in trickling filters.
- To remove the grease.

When chlorine is added to the treated sewage as a final step in the treatment is called post-chlorination. The post chlorination is carried out for the sewage having low bacterial count and will reduce the bacteria and BOD of the treated sewage.

When chlorine is added to the raw or partially treated sewage in order to control the bacterial growth is called pre-chlorination.

Pre chlorination is used to control the odour of sewage before it enters to the sedimentation tank, prevent the flies in trickling filter and helps to remove the grease in skimming units.

Chlorine may be added to the sewage at the beginning and end of the treatment process is called split chlorination.

ADVANCES IN SEWAGE TREATMENT

Advanced treatment of waste water is defined as the additional treatment needed to remove the suspended, colloidal and dissolved constituents remaining after conventional secondary treatment.

NEED FOR ADVANCED WASTEWATER TREATMENT:

- ❖ To remove the organic matter and TSS beyond what can be removed by secondary treatment.
- ❖ To remove nutrients, organic and inorganic constituents for re-use purposes.

TECHNOLOGIES USED FOR ADVANCED TREATMENT:

- ❖ Depth Filtration
- ❖ Adsorption
- ❖ Gas-stripping
- ❖ Ion-exchange
- ❖ Advanced oxidation process
- ❖ Distillation

DEPTH FILTRATION:

Depth filtration is an advanced wastewater treatment method, which involves the removal of particulate material suspended in sewage and also used to remove chemically precipitated phosphorous.

The various techniques available in depth filtration are,

- ❖ Deep bed down flow filters

- ❖ Deep bed up flow continuous backflow filters.
- ❖ Pulsed bed filters
- ❖ Travelling bridge filters
- ❖ Synthetic medium filters

ADSORPTION:

Adsorption is defined as the process of accumulating substances that are in solution on a suitable interface.

Adsorption is a mass transfer operation, in that a constituent in the liquid phase is transferred into solid phase.

The adsorbate is the substance that is being removed from the liquid phase at the interface.

The adsorbent is the solid, liquid or gas phase onto which adsorbate accumulates.

Activated carbon process is an example for the adsorption of advanced sewage treatment method.

GAS-STRIPPING:

Gas stripping involves the mass transfer of a gas from the liquid phase to the gas phase. The transfer is completed by contacting the liquid containing the gas that is to be stripped with a gas that does not containing any gas initially.

ION-EXCHANGE:

Ion-exchange is a unit process in which ions of a given species are displaced from an insoluble exchange material by ions of a different species in solution.

The most widely used method of ion-exchange is domestic water softening method.

Ion exchange method has been used in waste water application for the removal of nitrogen, heavy metals and TSS.

Ion exchange process can be operated in a batch or continuous mode.

In a batch process, the resin is stirred with water to be treated in a reactor, until the reaction is complete.

The used resin is removed by settling and simultaneously regenerated and reused.

In a continuous process, the exchange material is placed in a bed or a packed column and the water to be treated is passed through it.

ADVANCED OXIDATION PROCESS:

Advanced oxidation processes are used to oxidize the complex organic constituent found in wastewater, that are difficult to de-grade biologically into simpler end products.

When chemical oxidation is used, it may not be necessary to oxidize completely a given compound or group of compounds.

DISTILLATION:

Distillation is a unit operation in which the components of a liquid solution are separated by vaporization and condensation. Along with reverse osmosis, distillation can be used to control the formation of salts in critical reuse applications.

The method of distillation for wastewater reclamation is a recent development, the current development must be consulted for the results of efficient operations and more recent applications.